LOGOS CONSTRAINT ARCHITECTURE

Whitepaper v1.0 Blueprint (Scope Freeze)

Document Purpose: Lock architectural scope, mathematical core, and boundaries before full prose expansion.

Status: Frozen for white paper development

Date: 2026-02-28

SECTION 1: CORE CLAIMS (NON-NEGOTIABLE PILLARS)

1.1 Primary Claim

This framework is a **Normative Audit Framework for AI Stability**, not a universal invariant or moral authority system.

It enforces:

- Bounded option-space preservation
- Agency preservation (two-component metric)
- Explicit value declaration and versioning
- Recoverability guarantees with enforcement

1.2 What We Claim It Does

✓ Makes value assumptions visible instead of implicit ✓ Prevents parasitic optimization from collapsing user option-space ✓ Detects and logs drift in system behavior relative to declared constraints ✓ Provides measurable, auditable governance mechanisms ✓ Requires explicit override justification when constraints are breached ✓ Scales across domains through transparent value boundary declaration

1.3 What We Explicitly Do NOT Claim

✗ We do not solve alignment universally ✗ We do not eliminate the need for human judgment in governance ✗ We do not provide a moral authority or adjudicate between value systems ✗ We do not prevent all harms, only measurable option-space collapse ✗ We do not claim mathematical universality or culture-neutrality ✗ We do not solve the measurement problem for

option-space ✗ We do not address malicious actors who reject the framework entirely ✗ We do not eliminate the need for institutional accountability

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SECTION 2: ARCHITECTURAL LAYERS (FROZEN ORDER)

2.1 Critical Ordering Principle

RIC (Relational Invariance Constraint) must operate UPSTREAM of optimization and policy.

If RIC comes after weight optimization, it is not a constraint. It is a post-hoc filter.

Constraint boundaries have no power if they follow the decision.

2.2 The Stack (Final)

Layer 0: Base Model

- Generates candidate outputs/actions
- Operates under standard utility optimization logic
- No constraints applied at this level

Layer 1: Relational Invariance Constraint (RIC) — HARD FEASIBILITY GATE

- Evaluates all candidate actions against option-space and agency preservation bounds
- Three components (non-negotiable):
    - Agency Preservation
    - Epistemic Integrity
    - Recoverability
- Actions failing RIC are rejected immediately
- RIC cannot be weight-tuned or overridden by downstream optimization
- RIC is read-only with respect to the weight vector Λ

Layer 2: Logos Kernel (Stability Modules)

- Applies only to actions passing RIC
- Six submodules:
    - RCM (Relational Continuity Module)
    - NDOM (Non-Destructive Outcome Modeling)
    - PRM (Proportional Response Module)
    - EDM (Epistemic Dampening Module)
    - GBM (Generativity Bias Module)

    - APM (Autonomy Preservation Module)
- Reweights utilities among feasible actions
- Does not generate content; does not override policy

Layer 3: Policy & Compliance Layer

- Legal, platform, regulatory enforcement
- Can override RIC only with explicit logged justification
- Override triggers immediate review requirement
- Failed override predictions incur escalating penalties

Layer 4: Renderer

- Tone modulation
- Intensity scaling
- Epistemic calibration
- No constraint modification

2.3 Explicit Guarantee

Once RIC rejects an action, no downstream layer can resurrect it.

The constraint is binary and upstream.

SECTION 3: MATHEMATICAL CORE (FROZEN — NO ADDITIONS)

3.1 Five Core Equations (Only These)

Option-Space: O(s,H) = Set of reachable future states within horizon H Bounded by: physical feasibility, cognitive constraints, resource limits, social constraints

Viability (with Value Definition Layer):

V(s,H) = |O_viable(s,H)| Where O_viable(s,H) = O(s,H) - B B = explicitly declared boundary set of non-viable states B must be: enumerated, version-controlled, publicly auditable, justified

Agency (Two-Component):

A(s) = H(P(actions|s)) × C(s)

Where: H(P) = entropy of available meaningful action distribution C(s) = Changeability Coefficient (probability preference ordering can shift under reflective intervention)

Constraint: High entropy + C ≈ 0 = collapsed agency

Stability Constraint:

Action a is RIC-compliant if:

V_future(s,H) ≥ V_past(s,H) AND A_future(s) ≥ A_past(s)

Subject to Bounded Recoverability Rule (see Section 4)

Harm Tensor (Non-Collapsible):

$\vec{H}$ = (H_individual, H_collective, H_minority, H_future)
No single axis may dominate optimization without explicit justification. Imbalance detection monitors axis drift.

3.2 What We Do NOT Add

- No new constraint layers

- No new mathematical formalisms

- No additional stability metrics

- No extended harm dimensions

- No theoretical extensions

These five equations are the complete mathematical core.

SECTION 4: RECOVERABILITY & AGENCY METRICS

4.1 Bounded Recoverability Rule (Non-Negotiable)

Temporary contraction (V_future < V_past) is permissible ONLY if ALL conditions hold:

1. Finite Recovery Horizon:

H' ≤ H_max (H_max is domain-specific; see Section 6)

2. Intermediate Checkpoint:

$V(s_{H'/2}) \geq V(s_{past})$ Recovery must show progress at midpoint

Monotonic Improvement Trajectory:

$V(s_t)$ shows increasing trend for all t in $[H'/2, H']$

4. Failure Triggers:
- If recovery does not materialize: automatic rollback or constraint relaxation
- Logged penalty applied to system optimization
- Repeated failed predictions incur escalating weight penalties

5. Escalation Protocol:

Failed recovery → Review Required Repeated failures → Higher authority review Pattern of failures → Structural weight adjustment

4.2 Changeability Coefficient (C) — Operational Guidance

What C Measures:

Probability that an agent's preference ordering can shift under reflective or informational intervention.

How to Estimate (Non-Exhaustive):

- Behavioral attractor stability (how quickly does behavior return to baseline?)

- Reinforcement loop rigidity (do incentive changes shift behavior?)

- Longitudinal preference stability (do expressed preferences shift with new information over time?)

- Resistance to counterfactual information (does the agent update beliefs when encountering contradictions?)

Critical Constraint:

Measurement of C must not itself constrain agency.

Observational proxies preferred over interventional testing.

SECTION 5: DRIFT DETECTION & GOVERNANCE MECHANISMS

5.1 Independent Drift Monitoring (Required)

System must monitor, independently of reward U:

ΔV = Change in option-space viability ΔA = Change in agency preservation

These are monitored in parallel with, not subsumed into, the weight vector Λ.

Divergence Pattern:

If U increases while $\Delta V < 0$ or $\Delta A < 0$:

→ Proxy capture suspected

→ Triggers mandatory review

5.2 Weight Vector Protection

The weight vector $\Lambda = (\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ has hard lower bounds:

$\lambda_i \geq \lambda_{min}$ (floor, not tunable downward) All changes logged and version-controlled Multi-party approval required for shifts

5.3 Override Governance Protocol

RIC may only be overridden by:

1. **Legitimate governing authority** with explicit justification

2. **Reasoning fully logged** in system audit trail

3. **Immediate review trigger** (not deferred)

4. **Recovery prediction** with bounded H' and checkpoints

5. **Escalation** if recovery fails

Override is visible and costly, not silent and cheap.

SECTION 6: OPEN ENGINEERING QUESTIONS (NON-BLOCKING)

These are implementation challenges, **not conceptual gaps** in the framework.

6.1 Option-Space Approximation

Question: How to computationally approximate O(s,H) in real systems with large state spaces?

Scope: Numerical methods, graph algorithms, state abstraction strategies

Blocking: No — framework is valid even with approximate O

6.2 Changeability Coefficient Empirical Estimation

Question: What operational proxies best measure C(s) without manipulating autonomy?

Candidates:

- Longitudinal preference shift analysis

- Behavioral attractor stability metrics

- Belief update speed under new information

- Preference coherence over time

Blocking: No — framework scales with various C estimation methods

6.3 Domain-Specific H_max Values

Question: What recovery horizons are appropriate for different domains?

P**reliminary Table (Guidance Only):**

Domain	**H_max**	**Rationale**
Medical acute care	24-48 hours	Physiological recovery windows
Behavioral intervention	6-12 months	Standard intervention cycles
Addiction recovery	12-24 months	Neurological reset timescales
Systemic institutional reform	5-10 years	Organizational change timescales
Environmental remediation	10-50 years	Ecological recovery timescales

Blocking: No — each domain declares its own H_max explicitly

6.4 Recovery Penalty Scaling

Question: How to scale penalties for false recovery predictions?

Design Parameters:

- Escalating cost for repeated failures
- Magnitude of promised recovery (larger promise = larger penalty if wrong)
- Context factor (harm prevented vs. harm caused by contraction)

Blocking: No — framework is agnostic to penalty function, requires only transparency

6.5 Override Authority & Escalation Protocol

Question: What is the decision tree for authority levels overriding RIC?

Required Elements:

- Who can authorize override (individual, committee, institution)

- What evidence justifies override

- When does failed recovery escalate

- What triggers structural intervention

Blocking: No — framework requires only that overrides are logged and escalatable

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SECTION 7: HARM TENSOR EXTENSION

7.1 Multi-Axis Harm (Non-Collapsible)

Instead of scalar harm H, use vector:

$\vec{H}$ = (H_individual, H_collective, H_minority, H_future)

Definition:

- **H_individual:** Direct harm to identifiable agents

- **H_collective:** Distributed harm across populations

- **H_minority:** Disproportionate harm to underrepresented groups

- **H_future:** Harm to option-space of future agents (intergenerational)

7.2 Imbalance Detection

No single axis may dominate optimization without explicit justification:

If max($\vec{H}$) >> mean($\vec{H}$), flag for review

If optimization consistently deprioritizes one axis, log drift

7.3 What This Prevents

✗ Collapsing all harm into single metric (prevents political redefinition)

✗ Sacrificing minority interests for collective gain without visibility

✗ Discounting future harm in present optimization

✗ Hiding concentrated vs. distributed harm patterns

SECTION 8: IMPLEMENTATION ROADMAP (PHASES)

Phase 1: Formalization (**COMPLETE**)

- Option-space geometry ✓

- Agency metrics ✓

- RIC formulation ✓

- Drift detection architecture ✓

Phase 2: Domain-Specific Instantiation (**NEXT**)

- Declare value boundaries (B) for priority domains

- Specify H_max values with justification

- Define override authority structure

- Build penalty function

Phase 3: Computational Implementation

- Approximate O(s,H) algorithms
- C(s) estimation methods
- Drift detection implementation
- Audit logging infrastructure

Phase 4: Testing & Validation

- Stress-test on use cases
- Measure computational overhead
- Validate drift detection accuracy
- Test override escalation protocols

Phase 5: Governance & Deployment

- Institutional integration protocols
- Transparency & auditability standards
- Authority training and oversight
- Continuous monitoring and improvement

SECTION 9: LIMITATIONS & NORMATIVE TRANSPARENCY STATEMENT

9.1 What This Framework Cannot Do

- ***Does not eliminate measurement uncertainty***. Approximating O(s,H) and C(s) introduces error; the framework does not resolve this.

- ***Does not prevent malicious actors***. If institutional authority rejects the framework entirely, it has no power.

- ***Does not adjudicate between conflicting values***. The VDL makes values visible; it does not choose between them.

- ***Does not guarantee optimal outcomes.*** RIC constrains optimization; it does not guarantee the best result, only sustainable ones.

- ***Does not replace human judgment.*** Override authority, value boundary declaration, and domain-specific parameters all require deliberate human choice.

- ***Does not eliminate politics.*** Choosing H_max, penalty scaling, and authority structure are political acts. This framework makes them visible; it does not resolve them.

9.2 Normative Commitments (Explicit)

This framework encodes the following value assumptions:

1. ***Agency preservation matters.*** The capacity for self-authorship is treated as non-negotiable.

2. *Long-term viability over short-term gain.* The framework prioritizes sustainable option-space over immediate optimization.

3. *Transparency over efficiency.* Making constraints visible is valued over silent optimization.

4. *Measured over speculative recovery.* Checkpoints and penalties are preferred over abstract future promises.

5. *Distributed accountability*. Governance is multi-layered and requires explicit justification, not concentrated in single authority.

These are not discovered from nature. They are chosen.

Users of this framework inherit these commitments.

9.3 Domains of Uncertainty

- Measurement of option-space in high-dimensional systems

- Estimation of changeability without intervention

- Determination of "meaningful" vs. trivial actions

- Recovery timeline calibration across cultures

- Authority structure appropriate for different governance models

These are not solved in v1.0 and should not be claimed as solved.

SECTION 10: SCOPE BOUNDARY (WHAT IS EXCLUDED)

10.1 Out of Scope

- Detailed institutional governance models
- Specific penalty function parameterization
- Domain-specific value boundary libraries (beyond examples)
- Computational efficiency optimization
- Multi-agent equilibrium analysis
- Formal verification proofs
- Culture-specific fairness metrics
- Integration with specific AI platforms

10.2 In Scope (Core Deliverable)

- Architectural layer ordering and interaction
- Mathematical formulation of O, V, A constraints
- RIC as hard feasibility boundary
- Recoverability rule with enforcement
- Drift detection mechanisms
- Harm tensor structure
- Governance override protocol
- Implementation roadmap (phases 1-5)
- Explicit limitations and value commitments

SECTION 11: DOCUMENT FREEZE DECLARATION

Effective Date: 2026-02-28

Frozen Elements:

- ✓ Layer ordering (RIC upstream)
- ✓ Mathematical core (5 equations only)
- ✓ Core claims and explicit non-claims
- ✓ Recoverability rule structure
- ✓ Drift detection mechanisms
- ✓ Harm tensor formulation
- ✓ Open questions (engineering-level only)
- ✓ Scope boundaries

Not Frozen (Expansion Permitted):

- Implementation details within phases
- Specific penalty functions (form fixed, parameters tunable)
- Domain instantiations of H_max and VDL
- Authority protocols (structure fixed, instantiation variable)
- Computational methods (approaches, not specifics)

Next Step:

Full white paper prose expansion from this blueprint, maintaining all frozen elements verbatim.

No new concepts introduced.

No scope creep.

No metaphorical expansion beyond what serves clarity.

VERSION CONTROL

| Version | Date | Change | Status |

|---------|------|--------|--------|

| v1.0 Blueprint | 2026-02-28 | Initial scope freeze | **FROZEN |**

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